

ZAINAB IRFAN

AI/ML Engineer • RAG & LLM Systems Specialist • Full-Stack Python Developer

+92 324 8540177 | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL PROFILE

Results-driven AI/ML Engineer with hands-on, production-grade experience designing and shipping Retrieval-Augmented Generation (RAG) systems, conversational LLM agents, and full-stack AI platforms. Proven track record delivering end-to-end solutions — from data pipelines and model integration to deployed, customer-facing applications — using FastAPI, Next.js, LangChain, and vector databases such as Qdrant. Combines strong software engineering fundamentals with applied machine learning expertise, backed by 7 industry certifications from AWS, IBM, and DeepLearning.AI. Adept at translating business requirements into scalable, reliable AI products and comfortable owning projects independently in fast-paced, client-facing environments.

CORE TECHNICAL EXPERTISE

- **AI / LLM Engineering** — RAG Pipeline Architecture • LangChain • Prompt Engineering • LLM Fine-Tuning (LoRA/QLoRA, PEFT) • LLM Evaluation & Benchmarking • Conversational AI Agents • Streaming Chat Interfaces • Workflow Automation (n8n)
- **Machine Learning** — Supervised & Unsupervised Learning • Feature Engineering • Model Evaluation • Bias Detection • Exploratory Data Analysis (EDA)
- **Languages & Frameworks** — Python • JavaScript • HTML/CSS • FastAPI • Streamlit • Next.js • React • PyTorch • Transformers
- **Data & Vector Infrastructure** — Qdrant • Supabase • NumPy • Pandas • Scikit-learn • Matplotlib • Seaborn
- **APIs, Cloud & Deployment** — REST API Design & Integration • AWS (AI/ML) • Render Deployment • Infrastructure Automation • Google Colab (NVIDIA T4 GPU)
- **Languages Spoken** — English (Proficient) • Urdu (Native)

WORK EXPERIENCE

Freelance AI/ML Engineer & Full-Stack Developer

2024 – Present

Self-Employed / Independent Contractor

- Designed and built multiple production-style RAG platforms and LLM-powered applications, including document intelligence systems, conversational support agents, and workflow automation tools, using FastAPI, Next.js, LangChain, and Qdrant/Supabase.
- Owned the full software development lifecycle independently — API design, data pipeline development, model integration, cloud deployment (Render/AWS), and post-launch iteration — while managing client requirements and delivery timelines.
- Applied machine learning and data analysis techniques (EDA, feature engineering, model evaluation) to deliver predictive insights for clients across e-commerce and clinical/healthcare data use cases.
- Built automation systems (e.g., LLM-driven email response pipelines with n8n) that reduced manual operational workload for client businesses.

KEY PROJECTS

Prism AI — RAG-Based Document Intelligence Platform

2026

Stack: Next.js, FastAPI, LangChain, Qdrant, Supabase, OpenAI/Claude APIs • Deployed on Render

- Built a full-stack Retrieval-Augmented Generation platform that ingests multi-format documents (PDF, DOCX, PPT, CSV) and returns structured, context-aware answers.
- Designed the retrieval pipeline with LangChain and Qdrant for semantic search, using Supabase for persistent storage of documents and chat history.
- Integrated LLM APIs (OpenAI and Claude) with streamed chat responses for a real-time conversational Q&A experience.
- Built an analytics dashboard visualizing usage insights, queries per day, question volume, and engagement trends, with interactive charts.
- Deployed a production build with a Next.js frontend and a FastAPI backend hosted on Render.

Skills Applied: RAG Pipeline Design, LangChain, Vector Search (Qdrant), FastAPI, Next.js, LLM API Integration, Streaming Interfaces, Cloud Deployment, Data Visualization

Efficient LLM Fine-Tuning using LoRA & QLoRA (TinyLlama-1.1B-Chat)

2026

Stack: Python, PyTorch, Transformers, PEFT, TRL, BitsAndBytes, Hugging Face Datasets · Google Colab (NVIDIA T4 GPU)

- Fine-tuned TinyLlama-1.1B-Chat for Text-to-SQL generation using PEFT techniques, including LoRA and QLoRA, on ~3,000 curated examples from the Hugging Face b-mc2/sql-create-context dataset.
- Trained on Google Colab with an NVIDIA T4 GPU across 3 epochs, reaching 84.2% validation accuracy.
- Applied 4-bit quantization and low-rank adaptation to substantially cut GPU memory usage and training cost while preserving model performance.

Skills Applied: LLM Fine-Tuning, LoRA, QLoRA, PEFT, 4-bit Quantization, GPU Optimization, Hugging Face, PyTorch, Text-to-SQL

AI Customer Support Agent for Shopify Stores

2025 – 2026

Stack: Python, LLM APIs, Shopify Integration

- Developed an AI-powered customer support agent that integrates with a Shopify storefront to handle customer queries automatically.
- Used LLM APIs to generate context-aware responses, reducing manual reply workload for store operations.

Skills Applied: LLM API Integration, Conversational AI Agents, Third-Party API Integration, Python

Automated Email Reply System

2025

Stack: Python, LLM APIs, Email Automation, n8n

- Built an automation system that reads incoming emails, generates context-appropriate draft replies using an LLM, and sends responses automatically.

Skills Applied: Workflow Automation (n8n), LLM Prompt Engineering, Python, Email/API Integration

Comprehensive Clinical Data Analysis & Visualization

2025

Stack: Python, Pandas, Matplotlib/Seaborn, Scikit-learn

- Performed end-to-end exploratory data analysis (EDA) on raw clinical datasets, including cleaning, transformation, and feature engineering.
- Built visualizations of clinical insights such as patient distribution by disease and other key trends across the dataset.
- Applied machine learning techniques to generate predictive insights from patient records.

Skills Applied: Exploratory Data Analysis, Feature Engineering, Pandas, Data Visualization (Matplotlib/Seaborn), Scikit-learn, Predictive Modeling

AI PDF Question-Answering Platform

2025

Stack: Python, FastAPI, LLM APIs

- Built a tool that allows users to upload a PDF and ask natural-language questions, returning direct answers sourced from the document.

Skills Applied: FastAPI, Document Parsing, LLM API Integration, REST API Design

Machine Learning Models Development

2024 – 2025

- Trained 5+ classification and regression models with a full preprocessing pipeline: encoding, normalization, scaling, and train/test splitting.
- Evaluated models using accuracy, F1-score, precision/recall, and confusion matrices; selected optimal algorithms per dataset characteristics.

Skills Applied: Supervised & Unsupervised Learning, Model Evaluation, Scikit-learn, Feature Engineering

RESTful API Development & ML Integration

2024

- Engineered FastAPI-based REST APIs (GET/POST/PUT/DELETE) with request validation, integrating trained ML models as live prediction endpoints.

Skills Applied: FastAPI, REST API Design, ML Model Deployment, Request Validation

CERTIFICATIONS

- AI Fundamentals & Cloud — AWS
- AI Workflow: Feature Engineering & Bias Detection — IBM
- Building Systems with the ChatGPT API — DeepLearning.AI
- Machine Learning for All — University of London
- Machine Learning — Simplilearn
- App Deployment & Security — Packt

- [Learn to Code with AI — Scrimba](#)